CS19541 – COMPUTER NETWORKS LAB

PRACTICAL 10 – INTERNETWORKING IN CISCO PACKET TRACER

AIM

To set up internetworking with routers and a wireless router in Cisco Packet Tracer and check whether the connected devices can communicate.

REQUIREMENTS

Cisco Packet Tracer, routers, desktop PCs, a laptop, a wireless router, a cable modem, an Internet cloud, and a server.

PART A – ROUTER-BASED INTERNETWORKING

Configure Router1 to connect two separate networks. Assign 192.168.10.1/24 to FastEthernet0/0 and 192.168.20.1/24 to FastEthernet0/1. Set PC0 to 192.168.10.2 with default gateway 192.168.10.1, and PC1 to 192.168.20.2 with default gateway 192.168.20.1.

ROUTER COMMANDS

```
enable
configure terminal
interface fastethernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface fastethernet0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
end
```

PART B – WIRELESS ROUTER AND DHCP

Set the wireless router's network name (SSID) to HomeNetwork and turn on DHCP. Enter 208.67.220.220 as the DNS server. Connect the laptop through Wi-Fi and configure the PC to obtain an IP address automatically. On the server, set up a DHCP pool with starting address 208.67.220.1, subnet mask 255.255.255.0, DNS server 208.67.220.220, and a maximum of 50 users.

PROCEDURE

1. Add all required devices to the Cisco Packet Tracer workspace.
2. Connect the PCs to the router using straight-through Ethernet cables.
3. Set up the router interfaces and configure the IP settings on each PC.
4. Create the wireless network topology and configure its SSID and DHCP options.
5. Connect the laptop to the configured wireless network.
6. Test communication using ping or the Simple PDU tool.

INPUT

The input values include IP addresses, subnet masks, default gateways, the wireless SSID, and DHCP configuration details.

OUTPUT

Connected devices obtain the specified network settings and communicate through the configured networks. Successful connectivity can be checked with ping or Simple PDU.

OBSERVATION

The router interfaces were configured and activated. IP addresses and default gateways were assigned to the PCs. Wireless access and DHCP were configured, and network connectivity was tested.

RESULT

Therefore, internetworking with routers and a wireless router was configured and tested successfully in Cisco Packet Tracer.